





# Problems with DL models

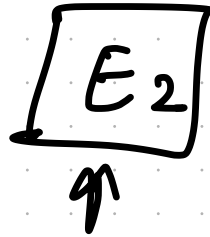
- Static knowledge
- Difficult updates → Niche topics
- Hallucinations

# Dense Passage Retriever

Contrastive learning



Query



Passage

# BART

- Denoising Transformer
- Can be easily finetuned for any task

# RAG Architecture

- Retriever
- Generator
- Vector DB
- Top K matches
- Prompt augmentation

# RAG $\neq$ Prompt Concatenation

- Prompt with one document at a time
- Conceptually run  $K$  such sessions
- Generate token probabilities
- Combine

$$p(y|x) = \sum_{z \in \text{Top-}K} p(z|x) \cdot P_{\Theta}(y|x, z)$$

$$\prod_i^N P_{\Theta}(y_i | x, z, y_{1:i-1})$$

## RAG Token: More customization

$$P(y|x) = \prod_i^N \sum_{z \in \text{Top-K}} P(z|x) \cdot P_\theta(y_i | x, z, y_{1:i-1})$$

- RAG-seq: Multiple decoding streams,  
combine them together
- RAG-token: Single decoding stream

# F;D: Forget about $P(z|x)$ (2021)

- Larger cross attention matrix
- Model learns  $P(z|x)$  through attention
- Small encoder, Large decoder



Quadratic  
self  
attention



Linear  
cross  
attention

- More documents, better performance

# ATLAS: Multitask RAG (2022)

- Retriever: Contriever  
↳ Unsupervised training
- Training over hundreds of tasks
- Few shot learning for any new task
- Small model RAG  $\geq$  Large model  
for a variety of tasks